RouteEd

End-to-End QA Report

Test execution: 28 October 2026 | Environment: React/Vite, Spring Boot, PostgreSQL 16, Docker, local Ollama llama3.2:3b

Outcome

Core learner and administrator flows passed after remediation. The reported assessment error was traced to a frontend timeout while the local model generated a route. The timeout was raised from 12 seconds to 90 seconds, and both starter diagnostics were expanded from two to six questions.

Area	Result	Evidence
Platform health	PASS	API health endpoint UP; PostgreSQL, Ollama, and API Docker services running.
Authentication	PASS	New learner and administrator registrations, login token, and /auth/me verified.
Admin builder	PASS	Created course, module, diagnostic, and question through protected APIs.
Learner diagnostic	PASS	Six-question Java diagnostic fetched; submission returned a score and AI path.
Progress and analytics	PASS	Module completion, 100% course progress, admin analytics, and one health score verified.
Build checks	PASS WITH WARNING	Frontend build/lint pass; Maven build passes but has no automated tests.

Key measured result

A complete custom course flow submitted in **5.56 seconds**, generated one personalized route module, recorded 100% progress after completion, and returned one analytics learner plus one health score. A Java diagnostic run measured 12.36 seconds before the timeout fix, validating the reported failure mechanism.

Issue evidence and remediation

The following screenshots were supplied during QA. They document the pre-fix UI symptoms, not the post-fix result.

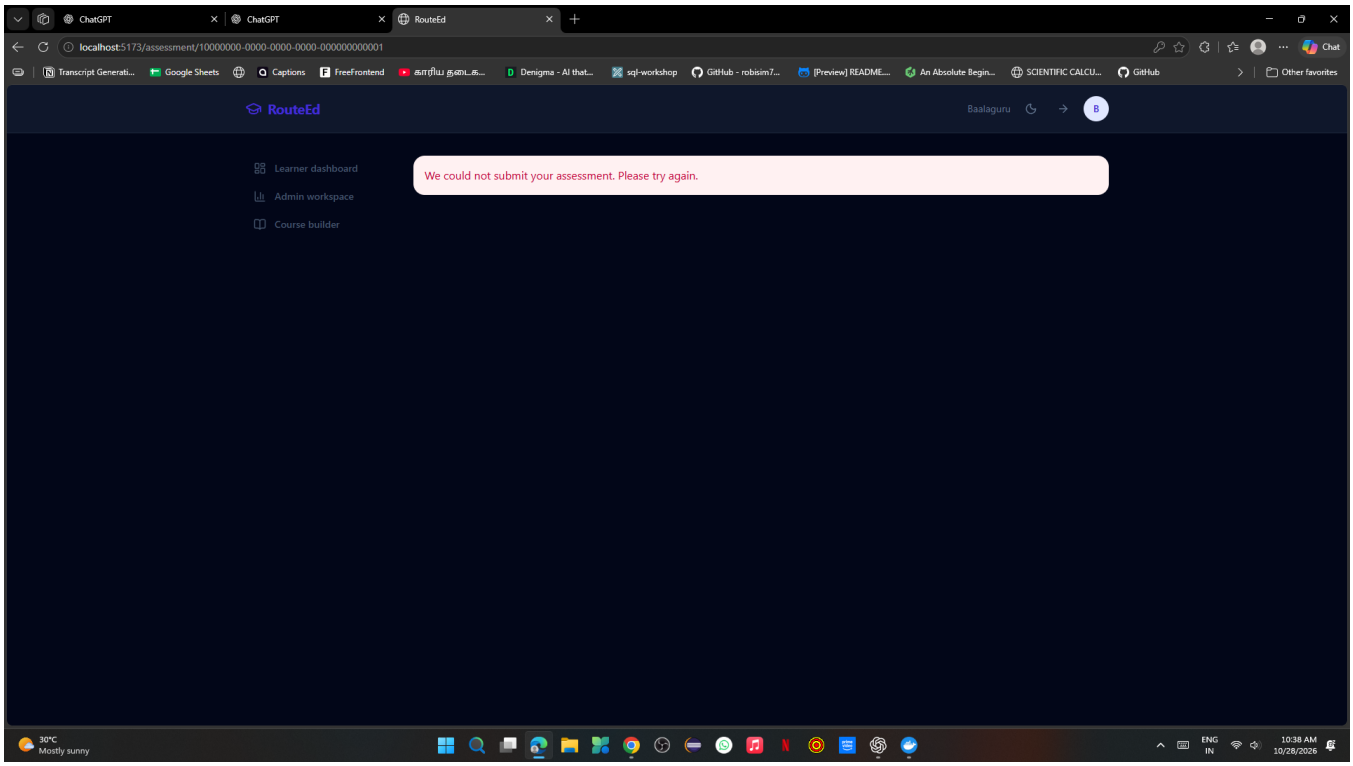


Figure 1. Assessment submission error supplied by the user. Root cause: the previous 12-second client timeout could expire while Ollama generated the personalized route.

Remediation applied

The Axios API client now waits up to 90 seconds for local AI-backed operations. Assessment errors also preserve an API-supplied message where available. This keeps the learner in the intended workflow instead of showing a false failure.

UI evidence and content coverage

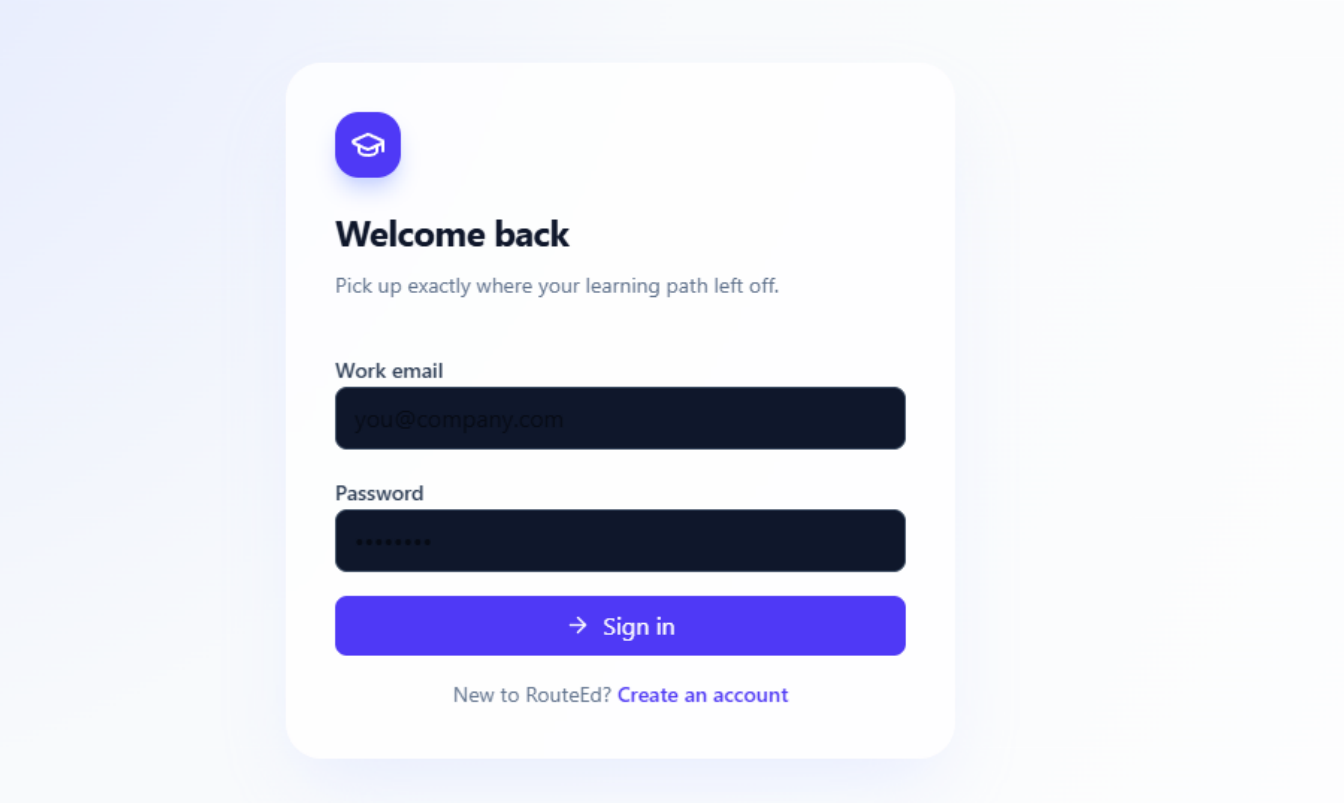


Figure 2. Login input contrast issue supplied by the user. The dark fields made entered values and placeholders difficult to read.

UI remediation

Input styles now define explicit slate text and placeholder colors for both light and dark themes. Java Foundations and Python Foundations each now have six diagnostic questions covering their respective skill tags.

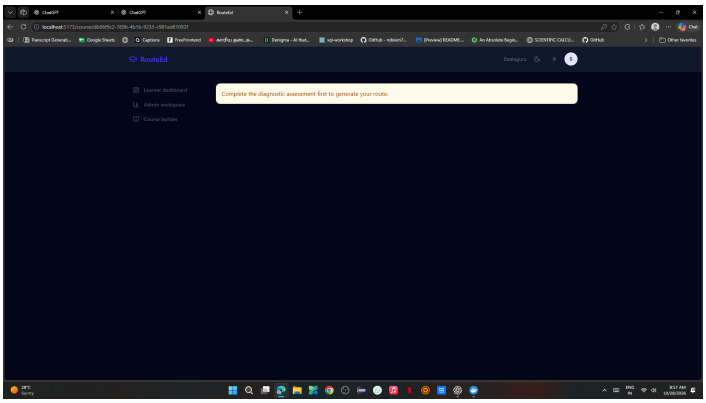


Figure 3. Learner workspace screenshot supplied during QA. The diagnostic call-to-action is visible, but the prior submission failure blocked route generation.

Open defects and recommended next actions

Nine tracked findings are recorded in bugs.md. Three were fixed during this QA pass. The highest-priority open items are listed below.

ID	Severity	Finding	Recommendation
BUG-004	Critical	Public registration accepts the ADMIN role.	Allow public learner signup only; use protected admin invite/provisioning.
BUG-005	High	Seeded Java/Python courses do not have an assigned admin, blocking their analytics.	Assign a seed admin or enforce organisation-level analytics permissions.
BUG-006	High	Maven reports no tests to run.	Add Spring Boot integration and security test coverage.
BUG-007	Medium	No test SMTP service is in Docker Compose.	Add Mailpit and surface delivery failures to admins.
BUG-008	Medium	Refresh-token rotation is not implemented.	Add rotation, revocation, and a client refresh interceptor.

QA limitations

Browser UI automation was unavailable in this workstation session, so visual findings use the supplied screenshots and source/build review. API end-to-end scenarios ran against the live local Docker stack. SMTP delivery was not sent because no development mail server or test mailbox is configured.

Deliverables

This PDF summarizes verification and findings. The companion bugs.md file contains reproduction details, impact, status, and recommended fixes.